Class 6 · Science

Chapter 1: The Wonderful World of Science

Class 6 Science · Ch 1 Worksheet B · Scientific Method Practice

Total marks: **15** · Questions: **7**

Instructions: Practise the steps of scientific inquiry. Write your answers in your notebook.

Q1. In the scientific method, after observation, the next step is to [1 mark]

- (A) forget the observation
- (B) ask a question about it
- (C) change the topic
- (D) write a long essay

Q2. A hypothesis is [1 mark]

- (A) a wild guess with no thought
- (B) an explanation that can be tested
- (C) the final answer
- (D) an opinion no one can check

Q3. In a fair test, you should change only _____ thing at a time. [1 mark]

Q4. Repeating an experiment is a waste of time once you have a result. [1 mark]

- (A) True
- (B) False

Q5. Write the steps of the scientific method in order. [3 marks]

Q6. Design a fair test to find out whether warm water dissolves sugar faster than cold water. Mention what you would change, keep the same, and measure. [4 marks]

Q7. Two students measure the height of the same plant. One says 28 cm; the other says 31 cm. Suggest two possible reasons for the difference and how to resolve it. [4 marks]

Answer Key

Class 6 Science · Ch 1 Worksheet B · Scientific Method Practice

Q1. In the scientific method, after observation, the next step is to

Answer: ask a question about it

After observation, scientists frame a focused question.

MCQ · EASY · 6-SCI-WOS-03

Q2. A hypothesis is

Answer: an explanation that can be tested

A hypothesis is a thoughtful, testable explanation.

MCQ · EASY · 6-SCI-WOS-03

Q3. In a fair test, you should change only _____ thing at a time.

Answer: one

Changing only one variable lets you connect the cause to the effect.

FILL_BLANK · MEDIUM · 6-SCI-WOS-04

Q4. Repeating an experiment is a waste of time once you have a result.

Answer: False

Repetition checks consistency and reduces the chance of error.

TRUE_FALSE · MEDIUM · 6-SCI-WOS-03

Q5. Write the steps of the scientific method in order.

Answer: Observe; ask a question; form a hypothesis; test the hypothesis with an experiment; record results; analyse; draw a conclusion; share findings.

Standard sequence of steps.

SHORT_ANSWER · MEDIUM · 6-SCI-WOS-03

Q6. Design a fair test to find out whether warm water dissolves sugar faster than cold water. Mention what you would change, keep the same, and measure.

Answer: Take two same-sized glasses, one with warm water and one with cold water of the same amount. Add the same spoonful of sugar to each at the same time and stir each at the same speed. Measure the time taken for the sugar to dissolve fully in each. Only the water temperature changes; everything else stays the same. Compare the two times.

A clear fair-test design with one variable changed.

SHORT_ANSWER · HARD · 6-SCI-WOS-04

Q7. Two students measure the height of the same plant. One says 28 cm; the other says 31 cm. Suggest two possible reasons for the difference and how to resolve it.

Answer: Possible reasons: (i) the scale was not placed at the very base of the plant, (ii) the plant was not held straight, or one student rounded their reading. Resolution: each student should measure again with the scale placed at soil level, the plant held vertical, and they should compare readings until they agree, taking the average of multiple readings.

Tests identifying measurement error sources and resolving them.

LONG_ANSWER · HARD · 6-SCI-WOS-04